Project Report: Real-Time Analysis of 3D Diffusion Fields

Subject: Multivariable Calculus & Control Systems Engineering

Project: Smart-Room Toxin Purifier v3

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1. Introduction

This project demonstrates the application of multivariable calculus and control theory in an environmental engineering context. We simulate a toxin concentration field defined as a function of four variables, ***C(x,y,z,t)***, and implement a "smart" purifier that uses real-time numerical integration to maintain air safety.

2. Mathematical Foundation

2.1 The 3D Diffusion Equation (PDE)

The movement of the toxin is governed by a second-order Partial Differential Equation:

$$\frac{\partial C}{\partial t} = D\left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2}\right)$$

Where D is the diffusion coefficient. To solve this on a computer, we use the ***Finite Difference Method***. The Laplacian (∇2) is calculated by measuring the concentration difference between a cell and its six neighbors.

Figure 1: The initial 3D grid setup showing the room in a "100% SAFE" state before toxin release.

2.2 Triple Integration for Total Mass

To provide feedback to the control system, the computer must calculate the total mass (T) of the toxin in the room at every time step. This is found using a triple integral over the room's volume V:

$$T(t) = \iiint_V C(x, y, z, t)\, dV$$

Numerical Implementation: This is solved as a ***3D Riemann Sum***. The code sums all elements in the 7×5×7 NumPy array and scales the result by the volume of a single grid cell (dV).

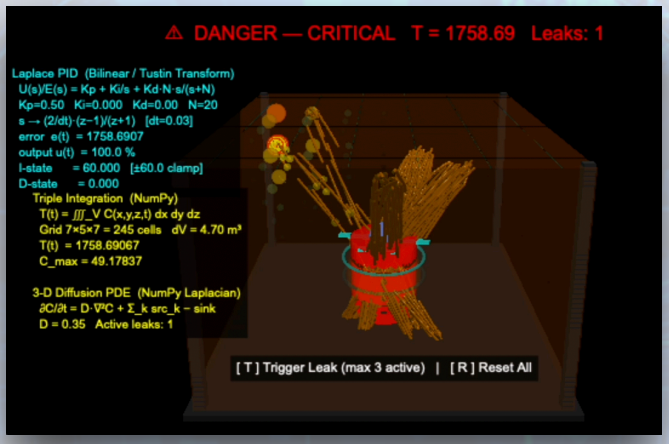

Figure 2: Active simulation during a leak. Note the calculated mass T = 1758.69, derived from the triple integral

3. Control Theory: Laplace Domain PID

The purifier's fan speed is not constant; it is regulated by a **PID Controller** designed in the **Laplace Domain (s)**. This allows the system to react to the rate of change and the accumulated history of the toxin levels.

Transfer Function:

$$H(s) = K_p + \frac{K_i}{s} + \frac{K_d \cdot N \cdot s}{s + N}$$

Digital Conversion: Because the simulation runs in discrete time steps (dt), we apply the

Tustin (Bilinear) Transform:

$$s \approx \frac{2}{dt}\frac{z-1}{z+1}$$

This mathematical conversion allows the Python script to execute complex continuous-time calculus using simple recursive arithmetic.

4. Data Analysis and Visualization

4.1 Vector Fields and Advection

The simulation visualizes the ***Gradient*** *(∇C)* using flow-field arrows. These arrows represent the vector field created by the purifier's suction, showing the path the toxin particles follow toward the filter.

4.2 Performance Graphs

The graphs at the bottom of the simulation track the performance of the math in real-time.

- **Red Line:** Represents the result of the **Triple Integral** (Total Mass).
- **Cyan Line:** Represents the **PID Output** (Fan Speed).

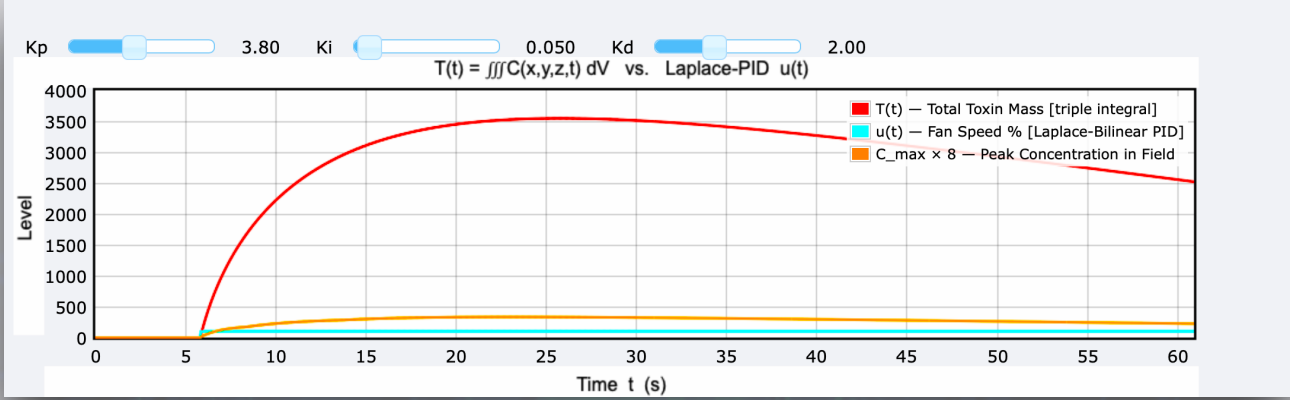

Figure 3: Live plot showing the controller successfully increasing fan speed to counteract a rising toxin integral

5. Conclusion

The project successfully bridges theoretical multivariable calculus and practical software engineering. By utilizing **NumPy** for vectorized matrix operations, we achieved the computational speed necessary to solve 3D partial differential equations and triple integrals in real-time.